

Domain 2 Study Module: Fundamentals of Generative AI

AWS Certified AI Practitioner (AIF-C01) Exam Preparation Program Domain Weight: 24% of scored content (~12 of 50 scored questions) Based on AIF-C01 Exam Guide Version 1.1 (April 30, 2026)

Part 1: Domain Introduction

What Is Generative AI, in Plain English?

Imagine you have a brilliant assistant who has read an enormous portion of the internet, books, articles, and code. This assistant cannot look anything up in real time and does not “know” things the way a database does. Instead, it has learned patterns in language, images, and other content so well that when you ask it a question or give it an instruction, it generates a brand-new response, word by word, based on what is statistically likely to come next given everything you have said so far.

That is generative AI (GenAI) in a nutshell: a category of artificial intelligence that creates new content (text, images, audio, video, code, or structured data) rather than simply analyzing existing content to produce a label or a number. When you chat with an AI assistant, ask it to write an email, or have it generate a product image, you are interacting with generative AI.

Domain 2 represents 24% of the scored exam content, roughly 12 of the 50 scored questions, making it the second-most heavily weighted domain after Domain 3. A candidate cannot pass the AIF-C01 exam without a solid grasp of these concepts, because Domain 2 vocabulary (tokens, foundation models, context windows, hallucinations) reappears constantly in Domains 3, 4, and 5.

How Domain 2 Builds on Domain 1

Domain 1 taught the vocabulary of traditional AI and ML: models that are trained to recognize patterns and then make a prediction, such as classifying an email as spam or predicting next month’s sales figure. These models are sometimes called **discriminative** or **predictive** models because they discriminate between categories or predict a specific value.

Domain 2 introduces a fundamentally different goal. Instead of analyzing input and returning a narrow answer (a category, a number, a yes/no), generative AI takes input and produces new, original output that did not exist before the request was made. This is the shift from “analyzing data” to “creating new content.”

	Domain 1 Mindset	Domain 2 Mindset
Core question	“What category does this belong to?” or “What number will this be?”	“What new content should be generated in response to this?”
Typical output	A label, score, or numeric prediction	Text, an image, audio, video, code, or structured data
Example	Predicting whether a transaction is fraudulent	Writing a draft customer service reply

What Makes Generative AI Fundamentally Different From Traditional ML

Three differences matter most for the exam:

1. **Output type.** Traditional ML produces a constrained, predictable output (a category or number). Generative AI produces open-ended, original content that can vary even when given the exact same input twice.
2. **Model reuse.** Traditional ML models are typically trained for one narrow task. Generative AI is built on foundation models: large, general-purpose models trained once on massive datasets and then adapted to many different tasks without retraining from scratch.
3. **Nondeterminism.** Traditional ML models given the same input will almost always produce the same output. Generative AI models can produce a different response each time, even with identical input, because the generation process involves an element of probability-based sampling.

Five Common Misconceptions Beginners Have About Generative AI

1. **“GenAI always makes things up.”** GenAI can hallucinate (state incorrect information confidently), but this is not the same as deliberately lying. It is a side effect of how the model predicts likely text, not an intentional choice. Techniques like Retrieval Augmented Generation (RAG) and grounding significantly reduce, though never fully eliminate, this risk.
2. **“Foundation models know everything, including current events.”** A foundation model only “knows” what was present in its training data up to its training cutoff. It has no live connection to the internet or company databases unless it is specifically connected to outside tools, search, or a knowledge base.
3. **“Bigger is always better.”** A model with more parameters is not automatically the best choice. Larger models often cost more, run slower, and may be unnecessary overkill for a simple task. Choosing the right-sized model for the use case is an exam-relevant skill.
4. **“Generative AI understands meaning the way a human does.”** A foundation model is a sophisticated pattern-matching and prediction system. It does not have beliefs, intentions, or true comprehension. It generates the statistically likely next piece of content based on patterns learned during training.
5. **“You need to be a data scientist to use generative AI well.”** Effective use of GenAI today is driven heavily by prompt engineering and the choice of managed services (like Amazon Bedrock), not by mathematical model-building. This is exactly the skill set the AIF-C01 exam validates.

Why a Non-Technical Professional Needs to Understand This

Generative AI is now embedded in everyday workplace tools: writing assistants, customer service chatbots, code completion tools, and image generators. A professional who understands the fundamentals can:

- Evaluate vendor claims about AI products with a critical eye instead of taking marketing language at face value.
- Participate meaningfully in decisions about which AWS GenAI services to adopt and why.
- Set realistic expectations with stakeholders about what GenAI can and cannot reliably do.
- Recognize early warning signs of risk, such as hallucination-prone use cases or unbounded cost exposure from token-based pricing.

Part 2: Official Task Statement Breakdown

Task Statement 2.1: Explain the basic concepts of generative AI (GenAI)

Full text: Explain the basic concepts of generative AI (GenAI).

Plain-English translation: Know the vocabulary that describes how generative AI works under the hood (tokens, embeddings, vectors, context, transformers) well enough to recognize and apply these terms in a business conversation, and understand the full lifecycle a foundation model goes through and the basics of agentic AI.

Specific learner skills: - *Define* tokens, chunking, embeddings, vectors, prompt engineering, transformer-based LLMs, foundation models, multi-modal models, and diffusion models. - *Identify* GenAI use cases across media types and business functions. - *Describe* the FM lifecycle from data selection through feedback collection. - *Describe* token-based pricing and its cost/performance impact. - *Describe* the role of context engineering. - *Define* foundational agentic AI concepts, including multi-agent systems, the Model Context Protocol (MCP), memory management, tool usage, and workflow orchestration.

Exam watch-out: Candidates frequently confuse “tokens” with “words.” A token is a chunk of text the model processes, which may be a whole word, part of a word, or punctuation. Questions that ask about cost or context window limits are really testing whether you know tokens drive both, not whether you can count words.

Realistic exam scenario stems: 1. “A company notices that its generative AI application is being charged more for longer customer support transcripts. Which concept BEST explains this cost behavior?” 2. “Which term describes the numerical representation of text that captures its meaning so that a model can measure similarity between pieces of content?” 3. “A development team wants to build an application where one AI agent retrieves data, a second agent summarizes it, and a third agent emails the result. Which concept describes this pattern?”

Task Statement 2.2: Understand the capabilities and limitations of GenAI for solving business problems

Full text: Understand the capabilities and limitations of GenAI for solving business problems.

Plain-English translation: Be able to weigh the pros and cons of generative AI honestly: know what makes it valuable to a business, know what can go wrong, know what factors should drive model selection, and know how a business should measure whether a GenAI investment is paying off.

Specific learner skills: - *Describe* the advantages of GenAI: adaptability, responsiveness, conversational capability, content generation. - *Identify* disadvantages: hallucinations, limited interpretability, inaccuracy, nondeterminism. - *Identify* factors for model selection: model type, performance requirements, capabilities, constraints, compliance, cost, latency, model complexity. - *Determine* business value and metrics: ROI, cross-domain performance, efficiency, conversion rate, average revenue per user, accuracy, customer lifetime value.

Exam watch-out: A common trap is an answer choice that lists a GenAI limitation as if it were a strength (for example, treating “nondeterminism” as a benefit because it produces “creative variety”). Read carefully whether the question is asking for an advantage or a disadvantage, since the exact same trait can be framed either way depending on the use case.

Realistic exam scenario stems: 1. “A legal department is concerned that an AI assistant might produce two different answers to the identical question asked twice. Which limitation of generative AI does this describe?” 2. “Which factor would be MOST important for a company selecting a foundation model for a real-time customer chat application with strict response-time requirements?” 3. “A retail company wants to measure whether its new GenAI product recommendation feature is delivering business value. Which metric would BEST demonstrate this?”

Task Statement 2.3: Describe AWS infrastructure and technologies for building GenAI applications

Full text: Describe AWS infrastructure and technologies for building GenAI applications.

Plain-English translation: Know which AWS services exist for building GenAI applications, understand why using AWS-managed services is usually smarter than building from scratch, and understand the cost levers involved in running GenAI workloads on AWS.

Specific learner skills: - *Identify* AWS services and features for GenAI development: Amazon Bedrock, Amazon SageMaker AI, SageMaker JumpStart, Amazon Quick, Kiro, Strands Agents, Amazon Bedrock AgentCore. - *Describe* the advantages of using AWS GenAI services versus building from scratch: accessibility, lower barrier to entry, efficiency, cost-effectiveness, speed to market, alignment with business objectives. - *Describe* the benefits of AWS infrastructure: security, compliance, responsibility, safety. - *Describe* cost tradeoffs: responsiveness, availability, redundancy, performance, regional coverage, token-based pricing, provisioned throughput, custom models.

Exam watch-out: Candidates often mix up Amazon Bedrock (using existing foundation models through a managed API) with Amazon SageMaker AI (building and training your own custom models). If a scenario describes a company that does NOT want to manage infrastructure or train its own model, the answer is almost always Amazon Bedrock.

Realistic exam scenario stems: 1. “A startup wants to add generative AI text summarization to its app without managing any servers or training any models. Which AWS service should it use?” 2. “A company expects unpredictable but high-volume usage of a foundation model and wants consistent, guaranteed response times. Which Amazon Bedrock pricing/capacity option addresses this need?” 3. “Which AWS offering is an open-source framework specifically intended for building AI agents capable of using tools and managing multi-step tasks?”

Part 3: Core Concept Notes - Domain 2 Deep Dive

3A. What Is Generative AI? (Building from Domain 1)

Generative vs. discriminative models. A discriminative (predictive) model, the kind covered in Domain 1, learns to draw boundaries between categories or to map inputs to a single output value. A generative model learns the underlying patterns and structure of its training data well enough to

produce brand-new examples that resemble that data. Put simply: discriminative models decide; generative models create.

Types of content generative AI can produce:

- **Text:** summaries, emails, articles, conversational replies, translations
- **Images:** illustrations, product mockups, design variations
- **Audio:** synthesized speech, music, sound effects
- **Video:** short clips, animations, video summaries
- **Code:** functions, scripts, unit tests, code explanations
- **Structured data:** synthetic tabular records for testing or augmenting datasets

“Generating” vs. “classifying” - why this changes everything. A classifier has a fixed, known set of possible outputs (spam or not spam, fraud or not fraud). A generator has an effectively unlimited space of possible outputs, because it is composing new content token by token or pixel by pixel. This means evaluation is harder (there is no single “correct” answer to compare against), creativity is possible, and so is hallucination, since the model can confidently generate something that sounds right but is not grounded in fact.

Real business examples of generative AI in use today:

- Customer service teams using AI assistants to draft responses to inbound support tickets
- Marketing teams generating first-draft ad copy and social media content
- Software teams using AI code generation tools to scaffold functions and write tests
- Retailers generating personalized product descriptions at scale
- Financial services firms summarizing lengthy regulatory documents for analysts

Limitations and failure modes every practitioner must know:

- Hallucination: confidently stated incorrect information
- Nondeterminism: different outputs for the same input
- Lack of true reasoning: pattern-based generation can fail at multi-step logic
- Bias: outputs can reflect biases present in training data
- Cost and latency: larger models cost more per token and can be slower
- Limited or no access to real-time information without external tools or RAG

3B. Foundation Models (FMs) and Large Language Models (LLMs)

What is a foundation model? A foundation model is a large AI model that is pre-trained on a massive, broad dataset and can then be adapted, with relatively little additional effort, to perform many different downstream tasks. Think of a foundation model like a culinary school graduate who has learned broad cooking fundamentals; that one chef can go on to specialize in pastry, sushi, or barbecue without starting culinary education over from scratch. The “foundation” has already been built.

Why FMs are trained once and adapted many times. Pre-training a model from scratch on a massive dataset requires enormous compute resources and time, often costing millions of dollars. Once that foundational work is done, the same base model can be reused across countless applications through prompt engineering, fine-tuning, or retrieval augmentation, which is dramatically cheaper and faster than starting over each time.

The relationship between FMs, LLMs, and generative AI. Generative AI is the broad umbrella concept (AI that creates content). A foundation model is the type of large, general-purpose, pre-trained model that powers most modern generative AI. A large language model (LLM) is a foundation model that specializes in understanding and generating human language (text). In other words: not every foundation model is an LLM (some generate images or audio), but every LLM is a foundation model, and both fall under the generative AI umbrella.

Pre-training in plain terms. Pre-training is the initial, large-scale training phase where a model learns general patterns of language, images, or other content by processing enormous volumes of data (text scraped from the web, books, code repositories, image collections, and so on). This phase requires massive compute infrastructure (often large clusters of specialized chips) and can take weeks or months. The output is a base model with broad general capability but no specialization in any one task.

What “billions of parameters” actually means. A parameter is one of the many internal numerical values a model adjusts during training to capture patterns in the data, similar to the countless small adjustments a sound engineer might make on a massive mixing board to get a song to sound right. A model with billions of parameters has billions of these adjustable values; more parameters generally means more capacity to capture complex patterns, but also more compute cost, more memory required, and typically slower and more expensive inference.

Multimodal models. A multimodal model can accept and/or generate more than one type of content, such as a model that can take both text and an image as input and return a text description, or a model that can take a text prompt and generate both an image and a caption. The exam expects awareness that multimodal capability is increasingly common in modern foundation models and is a relevant model-selection criterion.

Model providers available through Amazon Bedrock (awareness level):

Provider	What to Know for the Exam
Anthropic (Claude)	A leading LLM provider known for strong general-purpose reasoning and safety-focused design; available through Bedrock.
Meta (Llama)	An openly available family of LLMs; available through Bedrock for organizations wanting open-weight model options.
Mistral AI	A provider of efficient, competitively performing LLMs available through Bedrock.
Amazon (Titan and Nova)	Amazon’s own first-party foundation model families, built and offered directly by AWS through Bedrock.
Stability AI	A provider known for image-generation (diffusion) models, available through Bedrock.
Cohere	A provider focused on enterprise-oriented language models and embeddings, available through Bedrock.
AI21 Labs	A provider of LLMs focused on language understanding and generation, available through Bedrock.

The exam does not require deep technical comparison of these providers. What matters is recognizing that Amazon Bedrock’s value proposition is providing a single managed API to access multiple providers’ models, so an organization is not locked into one model and can choose the best fit for cost, performance, or compliance needs.

3C. How LLMs Work - The Non-Technical Explanation

Tokens. A token is the basic unit of text an LLM processes. A token is often, but not always, a whole word; it can be part of a word, a single character, or a punctuation mark. For example, “unbelievable” might be broken into multiple tokens such as “un,” “believ,” and “able.” Tokens matter because both the model’s context window limit and its pricing are measured in tokens, not words or characters.

Context window. The context window is the maximum amount of text (measured in tokens) that a model can “see” and consider at one time, covering both the input (prompt, conversation history, retrieved documents) and the output it generates. Think of it as the model’s short-term working memory. If a conversation or document exceeds the context window, older or excess content must be dropped, summarized, or excluded, and the model will not “remember” or use that excluded information when generating its response.

How an LLM predicts the next token. At each step, the model calculates a probability for every possible next token given everything that has come before, then selects one based on that probability distribution. This repeats one token at a time until the full response is generated. This is why the model can be described as “predicting” rather than “knowing”: it is making a statistically informed guess about what comes next, not retrieving a stored fact.

Temperature. Temperature is an inference parameter that controls how much randomness is allowed when the model selects the next token. Low temperature makes the model favor the most probable token consistently, producing more focused, predictable, conservative output. High temperature flattens the probability distribution, making less-likely tokens more competitive, which produces more varied, creative, and sometimes less coherent output.

Top-P and Top-K (plain-English). These are additional sampling controls that work alongside temperature: - **Top-K** limits the model to choosing its next token only from the K most probable candidate tokens (for example, the top 40 most likely options), discarding all lower-probability options entirely. - **Top-P** (also called nucleus sampling) limits the model to choosing from the smallest set of candidate tokens whose combined probability adds up to P (for example, 0.9, or 90%), which dynamically adjusts how many options are considered based on how confident the model is at that step.

Both parameters, like temperature, are levers for controlling the balance between predictable and creative output, and both are exam-relevant as “inference parameters.”

Why the same prompt can produce different answers. Because the next-token selection process involves probability-based sampling (influenced by temperature, Top-P, and Top-K), the exact same prompt can lead the model down a different sequence of token choices each time it runs, especially at higher temperature settings. This is the technical root of the nondeterminism discussed in Task Statement 2.2.

3D. Training Approaches for Foundation Models

Pre-training. The first and most resource-intensive stage. The model learns broad patterns of language, vision, or other content from an enormous, general-purpose dataset. This phase requires massive compute, can take weeks or months, and produces a general-purpose base model.

Fine-tuning. After pre-training, a model can be further trained on a smaller, more specific dataset to adapt it for a particular task or domain (for example, training a general LLM further on a company’s internal customer service transcripts so it adopts the right tone and terminology). Fine-tuning is far less expensive and time-consuming than pre-training because it builds on top of existing learned knowledge rather than starting over.

Reinforcement Learning from Human Feedback (RLHF). RLHF is a technique where human reviewers rate or rank model outputs (for example, choosing which of two responses is better), and that feedback is used to further train the model to produce outputs that align better with human preferences and values. This is a major reason modern LLMs feel more helpful, polite, and aligned with what users actually want compared to earlier raw pre-trained models.

Pre-training vs. fine-tuning vs. prompt engineering (overview):

Approach	What Happens	Cost Level
Pre-training	Model learns from scratch on a massive general dataset	Highest
Fine-tuning	Existing pre-trained model is further trained on a smaller, specific dataset	Moderate
Prompt engineering	No training occurs; the model’s existing knowledge is guided purely through carefully crafted input instructions	Lowest

Detailed coverage of prompt engineering techniques and full fine-tuning data-preparation steps belongs to Domain 3; Domain 2 only requires conceptual awareness of how these approaches relate to each other.

Instruction tuning. A specific type of fine-tuning where a model is trained on examples that pair an instruction with the desired response (for example, “Summarize this paragraph” paired with a correct summary). Instruction tuning is what makes a base model good at following user instructions rather than simply continuing text in a generic way.

3E. Hallucinations and Model Limitations

What hallucination is. A hallucination occurs when a generative AI model produces output that is factually incorrect, fabricated, or unsupported by any real source, but is presented in a fluent, confident tone that makes it sound true.

Why hallucinations happen. A foundation model does not “know” facts the way a database stores facts; it predicts the statistically likely next token based on patterns learned during training.

When the model has incomplete, ambiguous, or absent information relevant to a prompt, it can still generate a fluent-sounding continuation that fills the gap with plausible-but-incorrect content. This is not the model “lying,” since lying implies intent; it is an inherent side effect of how token prediction works.

Common hallucination types:

- **Factual errors:** stating an incorrect date, number, or fact with full confidence
- **Citation fabrication:** inventing a source, quote, or reference that does not exist
- **Logical inconsistencies:** contradicting earlier parts of the same response

How to detect and mitigate hallucinations (overview):

- **Retrieval Augmented Generation (RAG):** grounding responses in retrieved, verified company documents (detailed in Domain 3 and Domain 5)
- **Lower temperature settings:** reducing randomness to favor more conservative, likely-correct outputs
- **Human-in-the-loop review:** having a person verify high-stakes outputs before they are used
- **Output validation and confidence scoring:** automatically checking outputs against defined criteria (detailed in Domain 5)

Why hallucinations are a responsible AI concern (connection to Domain 4). Hallucinations directly threaten the responsible AI principle of veracity (producing truthful, accurate outputs) and can erode user trust, create legal exposure, and cause real-world harm if decisions are made based on fabricated information.

What practitioners should tell stakeholders about hallucination risk. Generative AI should not be treated as a guaranteed source of factual truth, especially for high-stakes decisions (medical, legal, financial). Outputs that matter should be grounded in verified data sources and, where consequences are significant, reviewed by a human before being acted upon.

3F. AWS Generative AI Services - Domain 2 Focus

Amazon Bedrock

- **One-line purpose:** A fully managed service that provides API access to foundation models from multiple providers without requiring customers to manage any infrastructure.
- **Primary use case:** Building GenAI applications (chatbots, summarization tools, content generators) quickly using existing pre-trained models.
- **Exam clue phrases:** “without managing infrastructure,” “access multiple foundation models through one API,” “no need to train a model from scratch.”
- **Commonly confused with:** Amazon SageMaker AI, which is for building and training custom models rather than using existing ones.
- **Realistic business scenario:** A bank wants to add an AI assistant that can answer customer questions about account features. It does not want to manage servers or train its own model, so it builds the assistant on Amazon Bedrock using an existing foundation model.

Amazon Bedrock Guardrails (overview; full detail in Domain 3)

- **One-line purpose:** A feature of Amazon Bedrock that lets organizations define content filters, topic restrictions, and safety boundaries to control what a foundation model is allowed to say or do.
- **Primary use case:** Preventing a GenAI application from generating harmful, off-topic, or noncompliant content.
- **Exam clue phrases:** “restrict topics,” “filter harmful content,” “enforce responsible AI policies on model output.”

Amazon Q (all variants)

Variant	Purpose	Exam Differentiator
Amazon Q Business	An AI assistant for business users that can answer questions and take action using an organization’s internal data and enterprise systems.	Look for “answer employee questions using company knowledge” or “search across enterprise content.”
Amazon Q Developer	An AI assistant focused on helping developers write, troubleshoot, and understand code, and answer questions about AWS services.	Look for “help developers code” or “explain AWS service usage.”
Amazon Q in Connect	An AI assistant embedded in Amazon Connect (contact center service) that helps customer service agents respond to customers in real time.	Look for “contact center agent assistance” or “real-time customer service guidance.”
Amazon Q in QuickSight (Amazon Quick)	Natural language query capability for business intelligence dashboards.	Look for “ask questions about data in plain language” or “BI dashboard.”

Amazon SageMaker AI JumpStart

- **One-line purpose:** A feature within SageMaker AI that provides a curated library of pre-built ML models and solution templates, including select foundation models, deployable with minimal setup.
- **Primary use case:** Quickly deploying or fine-tuning a pre-built model from within the SageMaker environment, for teams already working in SageMaker.
- **Exam clue phrases:** “curated model hub within SageMaker,” “deploy with minimal setup from SageMaker.”
- **Commonly confused with:** Amazon Bedrock, which is the dedicated managed FM platform with the broadest model selection, separate from the SageMaker interface.

Amazon Titan

- **One-line purpose:** Amazon’s own first-party family of foundation models, available through Amazon Bedrock, including Titan Text, Titan Embeddings, and Titan Multimodal/Image models.
- **Primary use case:** Organizations that want a foundation model built and supported directly by AWS, often for text generation, generating embeddings for RAG/vector search, or image generation/analysis.
- **Exam clue phrases:** “Amazon’s own foundation model,” “generate embeddings,” “Amazon-built model family.”
- **Commonly confused with:** Amazon Nova, which is also an Amazon-built model family available through Bedrock; both are first-party Amazon models, but the exam may simply expect recognition that Amazon offers its own FMs (Titan and Nova) alongside third-party providers.

PartyRock (Amazon Bedrock Playground - awareness level)

- **One-line purpose:** A no-code playground built on Amazon Bedrock that lets users experiment with building simple generative AI apps without writing code.
- **Primary use case:** Hands-on experimentation and learning, rather than production deployment.
- **Exam clue phrases:** “no-code experimentation with foundation models,” “playground for learning GenAI.”

Part 4: Key Terms and Definitions - Domain 2

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding
Generative AI	AI that creates new content (text, images, audio, video, code) rather than just analyzing existing data.	A chef inventing a new recipe rather than just identifying ingredients in a dish.	Core concept underlying the entire domain.	Confusing it with all of AI/ML; it is a subset focused on content creation.
Foundation Model (FM)	A large, pre-trained, general-purpose AI model that can be adapted for many tasks.	A culinary school graduate who can specialize in any cuisine after graduating.	Central concept tested across Domains 2 and 3.	Thinking an FM is built for one narrow task only.
Large Language Model (LLM)	A foundation model specialized in understanding and generating human language.	A widely read editor who can write in many styles and topics.	Tested alongside FM and GenAI definitions.	Treating “LLM” and “foundation model” as fully interchangeable terms.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding
Pre-Training	The initial large-scale training phase where a model learns general patterns from massive datasets.	Years of general schooling before choosing a specialty.	Differentiates from fine-tuning on the exam.	Believing pre-training happens quickly or cheaply.
Fine-Tuning	Further training a pre-trained model on a smaller, specific dataset to adapt it to a task or domain.	A general doctor completing a specialty residency.	Frequently tested cost/use-case comparison item.	Confusing fine-tuning with prompt engineering (no training occurs in prompting).
Instruction Tuning	A fine-tuning method that trains a model on instruction-response pairs so it follows directions well.	Teaching someone to follow recipe instructions exactly, not just cook freely.	Listed explicitly in Task Statement 2.1.	Thinking it is the same as RLHF.
RLHF	Training technique where human feedback on output quality is used to further improve a model.	A taste-testing panel guiding a chef's recipe adjustments.	Explains why modern models feel more aligned and helpful.	Believing it eliminates hallucinations entirely.
Token	A chunk of text (word, part of a word, or punctuation) that a model processes.	A puzzle piece; meaning comes from combining many pieces.	Drives both pricing and context window limits.	Assuming one token always equals one word.
Tokenization	The process of breaking text into tokens before a model processes it.	Cutting a sentence into puzzle pieces before assembling the picture.	Foundational mechanic behind token-based pricing.	Thinking tokenization only happens for English text.
Context Window	The maximum amount of text (in tokens) a model can consider at once.	The size of a desk; only so many papers can be spread out at one time.	Drives conversation length and document size limits.	Believing the model "remembers" everything from past conversations indefinitely.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding
Prompt	The input text or instruction given to a model to elicit a response.	A request you hand to a skilled assistant.	Foundation for all of prompt engineering (Domain 3).	Thinking only the literal question counts as “the prompt” (context and instructions count too).
Hallucination	Confidently stated, factually incorrect or fabricated output.	A storyteller who fills plot holes with confident, invented details.	Tested across Domains 2, 4, and 5.	Believing hallucination only happens with “bad” or low-quality models.
Temperature (in AI)	An inference parameter controlling randomness in next-token selection.	A thermostat for creativity: turn it up for spontaneity, down for consistency.	Common scenario-based question topic.	Thinking high temperature always improves output quality.
Top-P Sampling	A sampling method that selects from the smallest set of tokens whose combined probability reaches P.	Choosing from a shortlist that captures “most of the likely options,” sized dynamically.	Awareness-level inference parameter term.	Confusing it with Top-K, which uses a fixed count rather than a probability threshold.
Top-K Sampling	A sampling method that limits next-token choice to the K most probable candidates.	Choosing from a fixed top-10 list, regardless of how confident the model is.	Awareness-level inference parameter term.	Believing it behaves identically to Top-P.
Embedding	A numerical (vector) representation of text, images, or other data that captures meaning.	A GPS coordinate that places a concept’s “meaning” on a map.	Core to semantic search and RAG (Domain 3).	Thinking embeddings store the literal text rather than its meaning.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding
Vector	A list of numbers (often an embedding) representing data in a way a model can mathematically compare.	The numeric coordinates pinpointing a location.	Connects directly to vector databases.	Believing “vector” and “embedding” mean entirely different things (an embedding is a type of vector).
Latent Space	The mathematical space where embeddings/vectors live, where similar concepts sit close together.	A map where similar cities cluster near each other.	Helps explain why semantic search works.	Overcomplicating it as something only deep technical teams need to understand.
Multimodal Model	A model that can process and/or generate more than one type of content (text, image, audio).	A translator fluent in multiple “languages,” including pictures and sound.	Increasingly common model-selection criterion.	Assuming multimodal means “better” in every case.
Diffusion Model	A type of generative model, commonly used for images, that learns to generate content by reversing a process of gradually adding noise.	A sculptor revealing a statue by gradually removing material rather than building it up piece by piece.	Distinct model type tested alongside LLMs.	Believing diffusion models are used the same way as LLMs for text generation.
Text-to-Image	A generative AI capability that produces an image from a text description.	Describing a scene to an illustrator who then draws it.	Common use-case example on the exam.	Confusing it with text-to-text summarization tasks.
Text-to-Text	A generative AI capability that produces text output from text input (such as summarization or translation).	Asking an editor to rewrite or condense a document.	Most common GenAI use case category tested.	Assuming text-to-text only means direct translation.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding
Text-to-Code	A generative AI capability that produces program code from a natural language description.	Describing what you want built to a contractor who writes the blueprint.	Relevant to Amazon Q Developer use cases.	Believing it eliminates the need for any human code review.
Model Parameter	An internal numerical value a model adjusts during training to capture patterns.	A knob on a massive mixing board, adjusted during training to get the “sound” right.	Basis for understanding model size and complexity.	Equating “more parameters” automatically with “better outputs in every case.”
Model Weight	Another term for the learned parameter values stored inside a trained model.	The settings saved on that mixing board once the engineer is satisfied.	Relevant to “open-weight” model discussions.	Thinking “weight” refers to file size rather than learned values.
Inference	The process of a trained model generating a prediction or output for new input.	A chef cooking a dish using techniques already learned, rather than learning to cook for the first time.	Differentiates from training; tested across domains.	Confusing inference with training.
Training Data	The dataset used to teach a model patterns during pre-training or fine-tuning.	The cookbook and ingredients a chef studies and practices with.	Connects to bias and responsible AI discussions (Domain 4).	Assuming training data is always perfectly representative or unbiased.
Synthetic Data	Artificially generated data that mimics real data, often created by a generative model.	A flight simulator: realistic practice without using a real airplane.	Useful for filling data gaps; awareness-level term.	Believing synthetic data is always a perfect substitute for real-world data.
Zero-Shot Learning	A model performing a task correctly without being given any task-specific examples.	Asking someone to identify a fruit they have never directly been shown, using general knowledge alone.	Detailed further in Domain 3; introduced conceptually here.	Believing zero-shot prompts always produce the highest-quality result.

Term	Simple Definition	Everyday Analogy	Why It Matters for the Exam	Common Misunderstanding
Few-Shot Learning	A model performing a task after being shown a small number of examples within the prompt.	Showing someone two solved examples before asking them to solve a similar third problem.	Detailed further in Domain 3; introduced conceptually here.	Confusing few-shot prompting with fine-tuning (no training occurs).
Model Card	Structured documentation describing a model's intended use, limitations, and performance characteristics.	A nutrition label summarizing what is inside a packaged food product.	Connects directly to Domain 4 transparency tools (SageMaker Model Cards).	Thinking model cards are only relevant to highly technical audiences.
Model Provider	The company that builds and supplies a given foundation model (for example, Anthropic, Meta, Amazon).	A car manufacturer supplying engines for several different car brands.	Relevant to understanding Amazon Bedrock's multi-provider model.	Assuming AWS only offers its own first-party models.
API (in AI services)	A defined way for software to request a service (such as a model's output) over a network.	A restaurant menu: you submit an order in a defined format and receive a defined result.	Bedrock and most AWS GenAI services are described as "API-first."	Believing "API" is a complicated concept reserved for developers only.
Amazon Bedrock	AWS's fully managed service for accessing and building with foundation models from multiple providers.	A single universal remote that operates many different brands of devices.	One of the most heavily tested services in Domain 2 and 3.	Confusing it with SageMaker AI (custom model building).
Amazon Q	AWS's family of AI assistants for business users, developers, and other roles.	A team of specialized personal assistants, each suited to a different job function.	Tested for differentiating variants (Business, Developer, Connect).	Treating all Amazon Q variants as one single undifferentiated product.

Part 5: Comparison Tables - Domain 2

Table 1: Traditional ML vs. Generative AI

Aspect	Traditional ML	Generative AI
What It Does	Analyzes input to classify, predict, or detect patterns	Creates new, original content based on learned patterns
Output Type	A label, category, or numeric value	Text, images, audio, video, code, or structured data
Training Approach	Often trained for one narrow, specific task	Built on broad foundation models, then adapted for many tasks
Example	Predicting customer churn likelihood	Drafting a personalized customer outreach email
AWS Service Example	Amazon SageMaker AI (custom model training)	Amazon Bedrock (foundation model access)

Table 2: Foundation Model vs. Large Language Model vs. Diffusion Model

Aspect	Foundation Model	Large Language Model (LLM)	Diffusion Model
Definition	A large, general-purpose, pre-trained model adaptable to many tasks	A foundation model specialized in language understanding and generation	A generative model that creates content (often images) by reversing a noise-adding process
Primary Content Type	Varies (text, image, audio, multimodal)	Text	Primarily images (also used for some audio/video)
Relationship	Umbrella category that includes LLMs and diffusion models	A type of foundation model	A type of foundation model
Example Use Case	Powering a multimodal assistant	Powering a customer service chatbot	Generating a marketing image from a text prompt

Table 3: Pre-Training vs. Fine-Tuning vs. Prompt Engineering

Aspect	Pre-Training	Fine-Tuning	Prompt Engineering
Definition	Initial training of a model from scratch on a massive general dataset	Further training of an existing pre-trained model on a smaller, specific dataset	Crafting input instructions to guide an existing model's output, with no training involved

Aspect	Pre-Training	Fine-Tuning	Prompt Engineering
When to Use	Building an entirely new foundation model (rare for most organizations)	Adapting a model to a specific domain, tone, or task	Quickly guiding model behavior for a specific request without any training
Cost Level	Highest	Moderate	Lowest
AWS Support	Amazon SageMaker AI (custom training infrastructure)	Amazon Bedrock (fine-tuning select models) and SageMaker AI	Any Amazon Bedrock model, Amazon Q, or other GenAI interface

Table 4: Amazon Bedrock vs. Amazon SageMaker AI

Aspect	Amazon Bedrock	Amazon SageMaker AI
Purpose	Access and use existing pre-trained foundation models via managed API	Build, train, and deploy custom machine learning models from scratch
Who Uses It	Teams that want to build GenAI applications quickly without managing infrastructure	Data science and ML teams building bespoke models or needing full control over training
Model Access	Multiple third-party and Amazon-built foundation models through one API	No built-in foundation model marketplace by default (SageMaker JumpStart adds curated models)
Customization	Prompt engineering, RAG, and select fine-tuning of existing FMs	Full control over architecture, training data, and hyperparameters
Managed vs. DIY	Fully managed; no infrastructure to provision	More hands-on; requires configuring training jobs, endpoints, and compute

Table 5: Amazon Q Business vs. Amazon Q Developer vs. Amazon Lex

Aspect	Amazon Q Business	Amazon Q Developer	Amazon Lex
Purpose	AI assistant for business users answering questions using enterprise data	AI assistant for developers writing and troubleshooting code	Platform for building custom conversational chatbots and voice bots
Target User	Business and knowledge workers	Software developers	Application builders creating branded conversational interfaces

Aspect	Amazon Q Business	Amazon Q Developer	Amazon Lex
Key Capability	Searches and summarizes internal company knowledge and systems	Code generation, debugging help, and AWS service guidance	Natural language understanding for building custom dialogue flows
Exam Clue	“Answer employee questions using company data”	“Help developers write or fix code”	“Build a custom branded chatbot/voice bot for an application”

Table 6: High Temperature vs. Low Temperature Outputs

Aspect	Low Temperature	High Temperature
Output Style	Focused, consistent, predictable	Varied, creative, sometimes unexpected
Best Use Cases	Factual Q&A, code generation, structured data extraction	Creative writing, brainstorming, marketing copy ideation
Risk Level	Lower risk of incoherence, but may feel repetitive	Higher risk of incoherent or off-topic output

Table 7: In-Context Learning (Few-Shot) vs. Fine-Tuning vs. RAG (Overview)

Aspect	In-Context Learning (Few-Shot)	Fine-Tuning	RAG
Definition	Providing examples directly in the prompt to guide behavior, with no training	Further training the model on a specific dataset	Retrieving relevant external/company data at query time to ground the model’s response
Training Required	No	Yes	No (uses retrieval, not training)
Best For	Quick behavior adjustments for a single session or task	Long-term adaptation to a specific domain, tone, or specialized task	Keeping responses grounded in current, accurate, organization-specific information
Detail Level on Exam	Introduced here; tested in depth in Domain 3	Introduced here; tested in depth in Domain 3	Introduced here; tested in depth in Domain 3

Part 6: 25 Domain 2 Practice Questions

Beginner (Questions 1-10)

1. A company wants an AI system to write original marketing copy rather than classify customer reviews as positive or negative. Which type of AI capability does this require?

A. Predictive analytics B. Generative AI C. Computer vision D. Reinforcement learning

Correct Answer: B Rationale: Generative AI creates new, original content. Writing original marketing copy is a content-creation task, not a classification task.

2. Which term refers to the basic chunk of text that a large language model processes, which may be a whole word, part of a word, or punctuation?

A. Vector B. Token C. Parameter D. Embedding

Correct Answer: B Rationale: A token is the basic unit of text an LLM processes; it is not always a full word.

3. What is a foundation model?

A. A small model trained for exactly one narrow task B. A large, pre-trained, general-purpose model that can be adapted for many tasks C. A rules-based system with no learning involved D. A database that stores facts for retrieval

Correct Answer: B Rationale: A foundation model is defined by its broad, general-purpose pre-training and adaptability to many downstream tasks.

4. A foundation model confidently states an incorrect historical date as fact. What is this phenomenon called?

A. Overfitting B. Hallucination C. Underfitting D. Tokenization

Correct Answer: B Rationale: Hallucination is confidently stated, factually incorrect or fabricated output from a generative AI model.

5. Which AWS service provides managed access to foundation models from multiple providers through a single API, without requiring infrastructure management?

A. Amazon SageMaker AI B. Amazon Bedrock C. AWS Glue D. Amazon Redshift

Correct Answer: B Rationale: Amazon Bedrock is the managed service for accessing multiple providers' foundation models via API.

6. What does the "context window" of a large language model refer to?

A. The total lifetime storage capacity of the model B. The maximum amount of text the model can consider at one time C. The number of users who can query the model simultaneously D. The model's training dataset size

Correct Answer: B Rationale: The context window is the maximum amount of input and output text (measured in tokens) a model can consider during one interaction.

7. Which AWS AI assistant is designed specifically to help developers write, troubleshoot, and understand code?

A. Amazon Q Business B. Amazon Q Developer C. Amazon Lex D. Amazon Polly

Correct Answer: B Rationale: Amazon Q Developer is the variant of Amazon Q focused on coding and development tasks.

8. A model produces a slightly different answer each time it is given the exact same prompt. Which characteristic of generative AI does this illustrate?

A. Determinism B. Nondeterminism C. Overfitting D. Bias

Correct Answer: B Rationale: Nondeterminism describes the tendency of generative AI to produce varying outputs for identical inputs.

9. Which inference parameter controls how much randomness is introduced when a model selects its next token?

A. Context window B. Temperature C. Embedding D. Token limit

Correct Answer: B Rationale: Temperature directly controls the randomness/creativity of next-token selection.

10. What type of generative model is commonly used to create images by reversing a process of gradually adding noise?

A. Diffusion model B. Large language model C. Classification model D. Regression model

Correct Answer: A Rationale: Diffusion models generate content, typically images, by learning to reverse a noise-adding process.

Intermediate (Questions 11-20)

11. A company wants to deploy a generative AI chatbot but does not want to manage servers, choose between only one model provider, or build a model from scratch. Which AWS service BEST meets these requirements?

A. Amazon SageMaker AI B. Amazon Bedrock C. AWS Lambda D. Amazon EC2

Correct Answer: B Rationale: Amazon Bedrock provides managed, serverless access to multiple foundation model providers, matching all stated requirements.

12. A development team needs an AI assistant embedded into their contact center application to help live agents respond to customers in real time. Which Amazon Q variant is the BEST fit?

A. Amazon Q Business B. Amazon Q Developer C. Amazon Q in Connect D. Amazon Lex

Correct Answer: C Rationale: Amazon Q in Connect is purpose-built to assist contact center agents in real time within Amazon Connect.

13. A company is concerned about generative AI costs rising unexpectedly as customer conversations get longer. Which concept BEST explains why longer conversations increase cost?

A. Higher temperature settings B. Token-based pricing C. Increased model parameters D. Reduced context window

Correct Answer: B Rationale: GenAI services typically charge based on the number of tokens processed; longer conversations consume more tokens and increase cost.

14. A marketing team wants highly creative and varied ad copy ideas, even if some outputs are unconventional. Which inference parameter setting should they choose?

A. Low temperature B. High temperature C. Zero context window D. Minimum token count

Correct Answer: B Rationale: High temperature increases randomness and variety in output, which suits creative brainstorming tasks.

15. Which two of the following are advantages of generative AI for business use? (Select TWO.)

A. Guaranteed factual accuracy in every response B. Adaptability across many different domains and tasks C. Conversational, natural-language interaction capability D. Fully deterministic, identical output every time E. Elimination of all model bias

Correct Answer: B, C Rationale: Adaptability and conversational capability are recognized advantages of GenAI. The other options describe things GenAI does NOT reliably guarantee.

16. A company is choosing between two foundation models for a customer-facing chat application with a strict 2-second response time requirement. Which selection factor is MOST relevant here?

A. Model provider's headquarters location B. Latency C. Diffusion model availability D. Number of supported image formats

Correct Answer: B Rationale: Latency, the time it takes a model to generate a response, directly determines whether a response-time requirement can be met.

17. Which of the following correctly describes the relationship between embeddings and vectors?

- A. Embeddings and vectors are unrelated concepts
- B. An embedding is a type of vector that represents the meaning of data numerically
- C. A vector is always larger in size than an embedding
- D. Embeddings are only used for image data

Correct Answer: B Rationale: An embedding is a numerical vector representation that captures the meaning of text, images, or other data.

18. Order the following stages of the foundation model lifecycle in the correct sequence: (1) Fine-tuning, (2) Data selection, (3) Deployment, (4) Pre-training, (5) Evaluation.

- A. 2, 4, 1, 5, 3
- B. 4, 2, 1, 3, 5
- C. 1, 2, 4, 5, 3
- D. 2, 1, 4, 5, 3

Correct Answer: A Rationale: The FM lifecycle proceeds: data selection, pre-training, fine-tuning, evaluation, then deployment.

19. A company building a GenAI application wants to make sure the model only discusses approved topics and never generates offensive content. Which Amazon Bedrock feature directly supports this?

- A. Amazon Bedrock Knowledge Bases
- B. Amazon Bedrock Guardrails
- C. Amazon Bedrock Prompt Management
- D. Amazon SageMaker JumpStart

Correct Answer: B Rationale: Amazon Bedrock Guardrails allow organizations to define content filters and topic restrictions for model output.

20. Match each AWS GenAI offering to its BEST description.

Offerings: 1. Amazon Bedrock AgentCore, 2. Strands Agents, 3. Amazon Titan, 4. PartyRock

Descriptions: A. A no-code playground for experimenting with generative AI apps B. Amazon's own first-party family of foundation models C. A managed service for deploying and securing production-grade AI agents D. An open-source framework for building AI agents

Correct Matches: 1-C, 2-D, 3-B, 4-A **Rationale:** AgentCore is the managed production runtime for agents; Strands Agents is the open-source development framework; Titan is Amazon's first-party FM family; PartyRock is the no-code Bedrock playground.

Exam-Ready (Questions 21-25)

21. A pharmaceutical company is building a GenAI assistant to help researchers query internal lab reports. The company is extremely concerned about the assistant inventing chemical compound names that do not exist. Which combination of approaches would BEST reduce this risk?

- A. Increase temperature and remove all guardrails
- B. Use Retrieval Augmented Generation grounded in verified lab reports, combined with a lower temperature setting
- C. Switch to a diffusion model
- D. Increase the context window only, with no other changes

Correct Answer: B Rationale: Grounding responses in verified source documents (RAG) combined with a lower temperature to reduce randomness is the most direct way to reduce hallucination risk in a high-stakes domain.

22. A company is evaluating whether to adopt Amazon Bedrock instead of building and hosting its own foundation model infrastructure. Which of the following are valid business reasons to choose Bedrock? (Select TWO.)

A. Faster time to market B. Guaranteed zero hallucinations C. Lower barrier to entry without needing deep ML infrastructure expertise D. Complete elimination of all ongoing costs E. Automatic conversion of all generative output into structured data

Correct Answer: A, C Rationale: Speed to market and a lower barrier to entry are well-documented advantages of using managed AWS GenAI services rather than building from scratch. The other options are not realistic guarantees.

23. A retail company has deployed a GenAI product description generator. Six months later, leadership asks the AI team to demonstrate the business value of the investment. Which metric would be MOST appropriate to report?

A. Number of model parameters used B. Token-based pricing tier selected C. Return on investment (ROI) and conversion rate improvements D. Context window size in tokens

Correct Answer: C Rationale: ROI and conversion rate are business value metrics; the other options are technical configuration details, not business outcomes.

24. An organization wants to build a system where one AI agent searches internal documents, a second agent drafts a response, and a third agent reviews the draft for compliance issues before sending it, all coordinated automatically. Which Domain 2 concept BEST describes this overall pattern?

A. Single-shot prompting B. Multi-agent system / workflow orchestration C. Diffusion modeling D. Token-based pricing

Correct Answer: B Rationale: Multiple cooperating agents completing a coordinated, multi-step task is the definition of a multi-agent system using workflow orchestration.

25. A company needs its GenAI agent to call an external inventory management system to check stock levels before responding to a customer. Which concept describes the standardized way an AI agent connects to and calls external tools and systems?

A. Model Context Protocol (MCP) B. Diffusion model C. Token-based pricing D. Latent space

Correct Answer: A Rationale: The Model Context Protocol (MCP) is the standardized mechanism for connecting AI agents to external systems and tools.

Part 7: 30 Domain 2 Flashcards

Format: Front (term/question) / Back (definition/answer) / Exam Tip / Domain

Terminology Cards (10)

Card 1 Front: What is a token in the context of large language models? Back: The basic chunk of text an LLM processes; may be a full word, part of a word, or punctuation. Exam Tip: Tokens drive both pricing and context window limits, not words or characters. Domain: 2

Card 2 Front: What is a hallucination in generative AI? Back: Confidently stated, factually incorrect or fabricated output from a model. Exam Tip: Caused by token prediction patterns, not intentional deception; mitigated with RAG and lower temperature. Domain: 2

Card 3 Front: What is a context window? Back: The maximum amount of text (in tokens) a model can consider at one time, covering both input and output. Exam Tip: Exceeding the context window causes older content to be dropped or excluded. Domain: 2

Card 4 Front: What is a foundation model (FM)? Back: A large, pre-trained, general-purpose model that can be adapted to many different tasks. Exam Tip: “Foundation” implies broad capability built once, reused many times. Domain: 2

Card 5 Front: What does “temperature” control in an LLM? Back: The amount of randomness in next-token selection; low temperature is more focused/predictable, high temperature is more varied/creative. Exam Tip: Match low temperature to factual tasks, high temperature to creative tasks. Domain: 2

Card 6 Front: What is an embedding? Back: A numerical (vector) representation of text or other data that captures its meaning. Exam Tip: Embeddings enable semantic similarity comparisons, the foundation of vector search and RAG. Domain: 2

Card 7 Front: What is nondeterminism in generative AI? Back: The tendency for a model to produce different outputs given the exact same input. Exam Tip: A direct result of probability-based token sampling, not a bug. Domain: 2

Card 8 Front: What is the difference between pre-training and fine-tuning? Back: Pre-training builds a model from scratch on a massive general dataset; fine-tuning further trains an existing pre-trained model on a smaller, specific dataset. Exam Tip: Pre-training is the most expensive customization approach; fine-tuning is moderate cost. Domain: 2

Card 9 Front: What is Reinforcement Learning from Human Feedback (RLHF)? Back: A training technique where human ratings of model output quality are used to further improve the model’s alignment with human preferences. Exam Tip: RLHF is why modern LLMs feel more helpful and aligned than raw pre-trained models. Domain: 2

Card 10 Front: What is a multimodal model? Back: A model that can process and/or generate more than one type of content, such as text and images together. Exam Tip: Multimodal capability is a model-selection criterion, not automatically “better” in every scenario. Domain: 2

AWS Service Identification Cards (8)

Card 11 Front: Which AWS service provides managed API access to foundation models from multiple providers? Back: Amazon Bedrock. Exam Tip: If the scenario says “no infrastructure to

manage” and “access existing FMs,” the answer is Bedrock. Domain: 2

Card 12 Front: Which AWS AI assistant helps business users answer questions using enterprise data? Back: Amazon Q Business. Exam Tip: Distinguish from Amazon Q Developer (coding) and Amazon Q in Connect (contact centers). Domain: 2

Card 13 Front: Which AWS AI assistant helps developers write and troubleshoot code? Back: Amazon Q Developer. Exam Tip: Look for “developer,” “code,” or “AWS service questions” in the scenario. Domain: 2

Card 14 Front: Which AWS feature provides a curated library of pre-built models and solution templates inside SageMaker AI? Back: Amazon SageMaker JumpStart. Exam Tip: JumpStart lives inside the SageMaker interface, unlike Bedrock’s dedicated FM platform. Domain: 2

Card 15 Front: What is Amazon Titan? Back: Amazon’s own first-party family of foundation models (Text, Embeddings, Multimodal/Image), available through Bedrock. Exam Tip: Titan and Nova are both Amazon-built model families offered via Bedrock. Domain: 2

Card 16 Front: What is Strands Agents? Back: An open-source AWS framework providing building blocks for creating AI agents that use tools, manage memory, and complete multi-step tasks. Exam Tip: Strands Agents is the development framework; Amazon Bedrock AgentCore is the managed production runtime. Domain: 2

Card 17 Front: What is Amazon Bedrock AgentCore? Back: A managed service for building, deploying, securing, and scaling production-grade AI agents. Exam Tip: Pair with Strands Agents (build) vs. AgentCore (run in production). Domain: 2

Card 18 Front: What is PartyRock? Back: A no-code playground built on Amazon Bedrock for experimenting with generative AI applications. Exam Tip: Awareness-level only; associate it with “no-code experimentation.” Domain: 2

Comparison Cards (7)

Card 19 Front: What is the difference between fine-tuning and prompt engineering? Back: Fine-tuning further trains a model’s underlying parameters on new data; prompt engineering guides an existing model’s behavior through crafted input with no training involved. Exam Tip: If “no training occurs,” it is prompt engineering, not fine-tuning. Domain: 2

Card 20 Front: What is the difference between a foundation model and a large language model? Back: A foundation model is the broad category of large, general-purpose pre-trained models; an LLM is a foundation model specialized in language tasks. Exam Tip: Every LLM is an FM, but not every FM is an LLM (some are image- or audio-focused). Domain: 2

Card 21 Front: What is the difference between Amazon Bedrock and Amazon SageMaker AI? Back: Bedrock provides access to existing foundation models via managed API; SageMaker AI is for building and training custom models from scratch. Exam Tip: “Use an existing model” = Bedrock; “build/train our own model” = SageMaker AI. Domain: 2

Card 22 Front: What is the difference between Top-K and Top-P sampling? Back: Top-K limits next-token choice to a fixed number (K) of top candidates; Top-P limits choice to the smallest set of candidates whose combined probability reaches threshold P. Exam Tip: Top-K = fixed count; Top-P = dynamic, probability-based threshold. Domain: 2

Card 23 Front: What is the difference between Amazon Q Business and Amazon Lex? Back: Amazon Q Business is a ready-made AI assistant for employees using enterprise data; Amazon Lex is a platform for building custom-branded conversational bots for end-user applications. Exam Tip: “Build a custom chatbot for our app” = Lex; “Give employees an assistant for company knowledge” = Q Business. Domain: 2

Card 24 Front: What is the difference between Strands Agents and Amazon Bedrock AgentCore? Back: Strands Agents is the open-source framework used to build agents; AgentCore is the managed AWS service used to deploy, secure, and scale those agents in production. Exam Tip: Build vs. run; framework vs. managed runtime. Domain: 2

Card 25 Front: What is the difference between in-context learning (few-shot prompting) and fine-tuning? Back: In-context learning provides examples directly in the prompt with no training; fine-tuning permanently adjusts the model’s parameters through additional training. Exam Tip: In-context learning effects last only for that session/prompt; fine-tuning effects persist in the model itself. Domain: 2

Scenario Cards (5)

Card 26 Front: A company wants to add an AI assistant to its internal HR portal without managing any infrastructure. What should it use? Back: Amazon Q Business (paired with Amazon Bedrock if a custom application layer is also being built). Exam Tip: “Internal company knowledge” + “no infrastructure” strongly signals Amazon Q Business. Domain: 2

Card 27 Front: A company needs guaranteed, consistent response times for a high-volume production GenAI application on Amazon Bedrock. What capacity option addresses this? Back: Provisioned throughput. Exam Tip: Provisioned throughput reserves dedicated capacity for predictable performance, unlike on-demand pricing. Domain: 2

Card 28 Front: A legal team needs a model that avoids creative variation and produces the most consistent, predictable output possible for contract clause generation. What setting should be adjusted? Back: Lower the temperature (and/or Top-P / Top-K) inference parameter. Exam Tip: Low temperature = consistency; high temperature = creative variety. Domain: 2

Card 29 Front: A startup wants to build a custom-branded voice bot for its mobile app that understands natural language commands. What AWS service fits best? Back: Amazon Lex. Exam Tip: Lex is for building custom conversational interfaces, distinct from the ready-made Amazon Q assistants. Domain: 2

Card 30 Front: A company wants three AI agents to automatically collaborate: one to retrieve data, one to analyze it, and one to generate a report, with no human triggering each step. What concept and AWS offerings are relevant? Back: A multi-agent system with workflow orchestration, built using Strands Agents and deployed/managed with Amazon Bedrock AgentCore. Exam Tip: Multiple cooperating agents completing sequential steps automatically = multi-agent system / workflow orchestration. Domain: 2

Part 8: Domain 2 One-Page Cheat Sheet

Top 10 GenAI Concepts to Understand

1. Generative AI creates new content; traditional ML analyzes/classifies/predicts.
2. Foundation models are large, general-purpose, pre-trained models adapted for many tasks.
3. LLMs are foundation models specialized in language.
4. Tokens are the units models process; they drive both cost and context window limits.
5. The context window is the model's working-memory limit, measured in tokens.
6. Temperature, Top-P, and Top-K control how random/creative vs. focused/predictable outputs are.
7. Hallucination is confident, incorrect output, a side effect of token prediction, not intent.
8. Pre-training builds a model from scratch; fine-tuning adapts an existing model; prompting guides it with no training.
9. RLHF uses human feedback to align model behavior with human preferences.
10. Agentic AI involves agents that use tools, manage memory, and orchestrate multi-step workflows, often coordinated via the Model Context Protocol (MCP).

AWS GenAI Services at a Glance

Service	One-Liner
Amazon Bedrock	Managed API access to multiple foundation model providers
Amazon Bedrock AgentCore	Managed production runtime for AI agents
Strands Agents	Open-source framework for building AI agents
Amazon Q Business	AI assistant for employees using enterprise data
Amazon Q Developer	AI assistant for developers (code, AWS guidance)
Amazon SageMaker JumpStart	Curated pre-built model library inside SageMaker
Amazon Titan	Amazon's first-party foundation model family
PartyRock	No-code Bedrock playground for experimentation

Essential GenAI Vocabulary (One-Liners)

- **Token:** A chunk of text a model processes.
- **Context window:** Maximum text a model can consider at once.
- **Embedding:** A numeric vector representing meaning.
- **Hallucination:** Confidently wrong AI output.
- **Temperature:** Controls output randomness/creativity.
- **Fine-tuning:** Further training an existing model on specific data.
- **RLHF:** Human feedback used to align model behavior.
- **Multimodal model:** Handles more than one content type (text, image, audio).

Five Domain 2 Exam Watch-Outs

1. Do not equate "token" with "word."

2. Bedrock = use existing FMs; SageMaker AI = build/train custom models.
3. High temperature is not automatically “better”; match it to the use case.
4. Hallucination is not the same as bias; they are different responsible AI concerns.
5. Amazon Q has multiple variants; do not treat them as one interchangeable product.

Five “If the Question Says... Then the Answer Is...” Pairs

1. If the question says “access multiple FM providers without managing infrastructure,” then the answer is Amazon Bedrock.
 2. If the question says “help developers write/debug code,” then the answer is Amazon Q Developer.
 3. If the question says “guaranteed consistent capacity for high-volume production traffic,” then the answer is provisioned throughput.
 4. If the question says “open-source framework to build agents,” then the answer is Strands Agents.
 5. If the question says “no-code experimentation with generative AI,” then the answer is Party-Rock.
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Part 9: Domain 2 Summary and Review Plan

Five Key Domain 2 Takeaways

1. Generative AI creates new content using foundation models; this is fundamentally different from the predictive, classification-focused mindset of Domain 1.
2. Tokens, context windows, and inference parameters (temperature, Top-P, Top-K) govern both the cost and the behavior of generative AI applications.
3. Hallucination and nondeterminism are inherent characteristics of how generative AI works, not signs of a “broken” model; they must be actively managed.
4. Amazon Bedrock is the central AWS service for accessing foundation models without managing infrastructure, while Amazon Q provides ready-made AI assistants for specific roles and use cases.
5. Agentic AI concepts (multi-agent systems, MCP, tool usage, workflow orchestration) are an increasingly important and newly emphasized part of this domain.

Remediation Plan for Learners Scoring Below 60%

1. Re-read Part 3A and 3B to firmly separate “generative AI” from “traditional ML” before anything else.
2. Build a personal one-sentence definition for each of these eight terms before moving on: token, context window, foundation model, LLM, embedding, hallucination, temperature, fine-tuning.
3. Redo the Part 6 practice questions and write a one-sentence explanation, in your own words, for every question answered incorrectly.
4. Focus extra review time on the Amazon Bedrock vs. SageMaker AI distinction and the Amazon Q variant differences in Part 5, since these are the most commonly confused exam topics in this domain.
5. Retake the Part 6 question set after 24 hours to confirm retention before moving on to Domain 3.

3-Day Domain 2 Review Schedule

Day 1: Re-read Part 1 (Introduction) and Part 3 (Core Concept Notes) in full. Self-quiz using the Part 4 term table by covering the definition column and trying to recall each term from memory.

Day 2: Work through all seven Part 5 comparison tables and the Part 7 flashcard deck (30 cards). Identify any card you hesitate on and flag it for a second pass.

Day 3: Complete the full 25-question Part 6 practice set under light time pressure (aim for under 25 minutes). Review every incorrect answer's rationale, then do a final pass of the Part 8 cheat sheet the night before moving to Domain 3.

How Domain 2 Concepts Feed Into Domains 3, 4, and 5

- **Domain 3 (Applications of Foundation Models):** Builds directly on tokens, context windows, temperature, and fine-tuning concepts introduced here, applying them to prompt engineering, RAG, and FM evaluation in much greater depth.
- **Domain 4 (Guidelines for Responsible AI):** Builds on the hallucination and bias awareness introduced in Part 3E, extending it into formal responsible AI principles, fairness, and AWS tools like SageMaker Clarify.
- **Domain 5 (Security, Compliance, and Governance):** Builds on the agentic AI and AWS infrastructure concepts introduced here, extending them into securing AI agents (Amazon Bedrock AgentCore Identity), grounding techniques for hallucination mitigation, and governance frameworks.

End of Domain 2 Study Module: Fundamentals of Generative AI Document formatted for conversion to Word (.docx) and PDF deliverables AIF-C01 Exam Guide Version 1.1 (April 30, 2026) | Domain Weight: 24% (~12 scored questions)